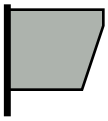
Math Kangaroo Practice - Grades 3-4

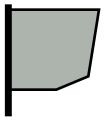
Practice set for 2002 #15
10 practice problems per selected problem.

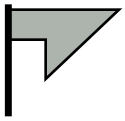
Set 1 - Flag Silhouette in the Wind

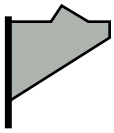
Based on 2002, Problem 15

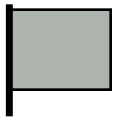
Practice 1.1 Elena watches a rectangular flag attached along its left vertical edge to a pole and sketches 5 silhouettes as the wind blows. Which sketch **cannot** happen unless the flag is torn from the pole?

A) 

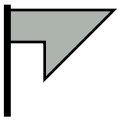
B) 

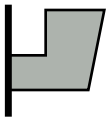
C) 

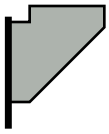
D) 

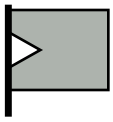
E) 

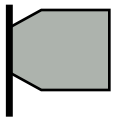
Practice 1.2 Four of the 5 flag silhouettes below are impossible unless the rectangular flag is torn from the pole. Which silhouette **can** happen with an intact flag?

A) 

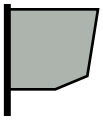
B) 

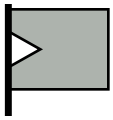
C) 

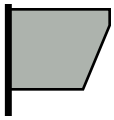
D) 

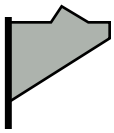
E) 

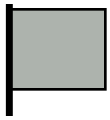
Practice 1.3 Maya watches a rectangular banner attached along its left vertical edge to a pole and sketches 5 silhouettes as the wind blows. Which sketch **cannot** happen unless the banner is torn from the pole?

A) 

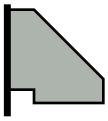
B) 

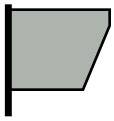
C) 

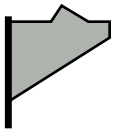
D) 

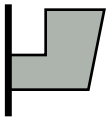
E) 

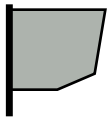
Practice 1.4 Sam watches a rectangular flag attached along its left vertical edge to a pole and sketches 5 silhouettes as the wind blows. Which sketch **cannot** happen unless the flag is torn from the pole?

A) 

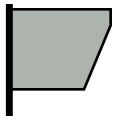
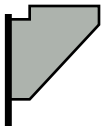
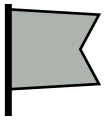
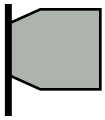
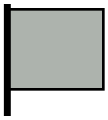
B) 

C) 

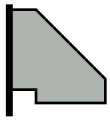
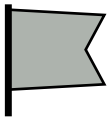
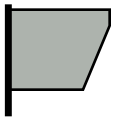
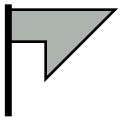
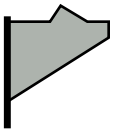
D) 

E) 

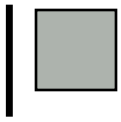
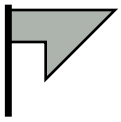
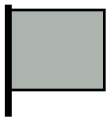
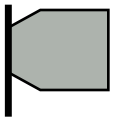
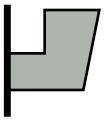
Practice 1.5 Maya watches a rectangular flag attached along its left vertical edge to a pole and sketches 5 silhouettes as the wind blows. Which sketch **cannot** happen unless the flag is torn from the pole?

A)  **B)**  **C)**  **D)**  **E)** 

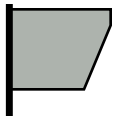
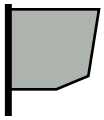
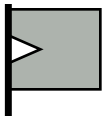
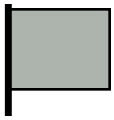
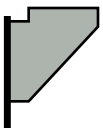
Practice 1.6 Sam watches a rectangular banner attached along its left vertical edge to a pole and sketches 5 silhouettes as the wind blows. Which sketch **cannot** happen unless the banner is torn from the pole?

A)  **B)**  **C)**  **D)**  **E)** 

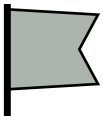
Practice 1.7 Four of the 5 flag silhouettes below are impossible unless the rectangular flag is torn from the pole. Which silhouette **can** happen with an intact flag?

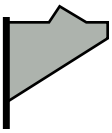
A)  **B)**  **C)**  **D)**  **E)** 

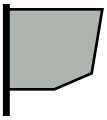
Practice 1.8 Sam watches a rectangular flag attached along its left vertical edge to a pole and sketches 5 silhouettes as the wind blows. Which sketch **cannot** happen unless the flag is torn from the pole?

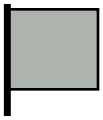
A)  **B)**  **C)**  **D)**  **E)** 

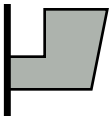
Practice 1.9 Elena watches a rectangular banner attached along its left vertical edge to a pole and sketches 5 silhouettes as the wind blows. Which sketch **cannot** happen unless the banner is torn from the pole?

A) 

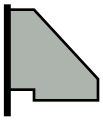
B) 

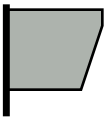
C) 

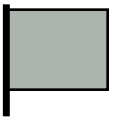
D) 

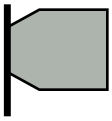
E) 

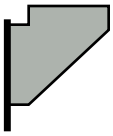
Practice 1.10 Elena watches a rectangular flag attached along its left vertical edge to a pole and sketches 5 silhouettes as the wind blows. Which sketch **cannot** happen unless the flag is torn from the pole?

A) 

B) 

C) 

D) 

E) 

Answer Key & Explanations

For generated/2026-10-19-21-32-11.pdf (Practice set for 2002 #15)

Solutions & kid-friendly explanations

1.1: C Because the left edge of the rectangular flag is fastened to the pole from top to bottom of the attachment, the silhouette must touch the entire attachment segment along the pole. Picture C is missing the lower part along the pole, which can only happen if the flag is torn.

1.2: C Only in C is the flag attached continuously along its full vertical segment on the flagpole. In A, B, D, E, parts of the flag are detached from the pole.

1.3: B In picture B, there is a V-shaped gap right along the middle of the flagpole where the flag is attached. Since the left edge is fixed to the pole, that gap cannot appear unless the flag is ripped.

1.4: D Because the left edge of the rectangular flag is fastened to the pole along the entire attachment segment, the silhouette must cover that full segment on the pole. Picture D is missing the upper half along the pole, which is impossible unless the flag is torn.

1.5: D In picture D, both the top and bottom corners of the flag's left edge have pulled away from the flagpole. That cannot happen unless the flag is torn at both ends of its attachment.

1.6: D Because the left edge of the rectangular flag is fastened to the pole from top to bottom of the attachment, the silhouette must touch the entire attachment segment along the pole. Picture D is missing the lower part along the pole, which can only happen if the flag is torn.

1.7: C Only in C is the flag attached continuously along its full vertical segment on the flagpole. In A, B, D, E, parts of the flag are detached from the pole.

1.8: C In picture C, there is a V-shaped gap right along the middle of the flagpole where the flag is attached. Since the left edge is fixed to the pole, that gap cannot appear unless the flag is ripped.

1.9: E Because the left edge of the rectangular flag is fastened to the pole along the entire attachment segment, the silhouette must cover that full segment on the pole. Picture E is missing the upper half along the pole, which is impossible unless the flag is torn.

1.10: D In picture D, both the top and bottom corners of the flag's left edge have pulled away from the flagpole. That cannot happen unless the flag is torn at both ends of its attachment.